

BRIDGEPORT SIKORSKY, CT, USA

WMO: 725040

Lat: 41.164N

Lon: 73.127W

Elev: 2

StdP: 101.31

Time Zone: -5.00 (NAE)

Period: 96-19

WBAN: 94702

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.8	-9.3	-22.1	0.5	-9.9	-19.5	0.7	-7.5	12.2	-0.6	11.1	-0.8	4.9	350	0.512

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.8	31.3	22.9	29.7	22.2	28.3	21.6	24.6	28.7	23.9	27.4	23.2	26.4	4.9	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.4	18.2	26.7	22.8	17.5	26.0	22.2	16.8	25.4	74.3	28.8	71.5	27.5	68.9	26.4	27.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.8	9.2	8.3	-15.0	34.7	3.0	2.0	-17.2	36.2	-18.9	37.3	-20.5	38.5	-22.7	39.9
			-16.1	26.0	2.6	0.8	-18.0	26.5	-19.6	27.0	-21.1	27.4	-23.0	28.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.8	-0.5	0.7	4.2	9.9	15.3	20.6	23.9	23.4	19.7	13.4	7.6	2.8		
	DBStd	9.37	5.16	4.43	4.30	3.73	3.56	3.33	2.68	2.65	3.34	4.09	4.18	4.51		
	HDD10.0	1157	327	262	186	46	1	0	0	0	0	14	94	227		
	HDD18.3	2885	586	495	438	254	108	16	0	1	23	159	322	482		
	CDD10.0	1818	1	1	6	42	167	318	431	416	292	121	22	3		
	CDD18.3	503	0	0	0	1	16	84	172	158	65	8	0	0		
	CDH23.3	3064	0	0	1	11	82	456	1192	1021	283	18	0	0		
(13)	CDH26.7	732	0	0	0	2	15	108	334	233	38	2	0	0		
(14)	Wind	WSAvg	3.9	4.3	4.2	4.5	4.3	3.9	3.5	3.4	3.7	4.0	4.0	4.1		
(15) Precipitation	PrecAvg	1056	79	74	98	94	99	83	91	93	88	76	90	86		
	PrecMax	1878	284	169	234	272	242	450	326	213	188	178	260	200		
	PrecMin	585	10	11	18	18	10	10	12	18	14	8	9	21		
	PrecStd	238	52	34	47	51	51	77	66	52	51	40	53	46		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.7	13.5	18.6	24.7	28.6	31.9	33.9	32.9	30.0	25.7	19.1	15.7		
		MCWB	10.8	8.8	11.6	14.8	19.4	22.5	23.9	24.3	22.8	19.9	16.0	12.2		
	2%	DB	10.1	10.5	14.8	20.5	25.6	29.5	31.7	30.5	27.6	23.0	17.2	13.2		
		MCWB	7.8	7.4	9.3	12.6	18.1	21.1	23.0	23.0	21.3	19.4	14.1	11.2		
	5%	DB	8.2	8.4	12.1	17.9	23.2	27.7	30.0	29.1	26.2	21.3	15.9	11.3		
		MCWB	5.8	5.7	8.0	11.4	16.8	20.5	22.5	22.3	20.8	17.8	13.1	9.7		
	10%	DB	6.5	6.7	10.1	15.8	21.1	26.0	28.4	27.8	24.9	19.8	14.3	9.6		
		MCWB	4.4	4.1	6.5	10.2	15.6	19.7	21.9	21.7	20.3	16.4	11.8	7.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.7	10.8	13.4	17.1	21.2	23.9	25.6	25.7	24.5	21.9	17.4	13.7		
		MCDB	12.4	11.9	17.0	22.1	26.3	29.2	31.6	30.5	27.7	23.5	18.9	15.1		
	2%	WB	8.6	8.1	10.7	14.7	19.7	22.8	24.6	24.7	23.4	20.3	15.4	11.6		
		MCDB	9.6	9.7	13.4	18.1	23.8	27.3	29.3	28.3	25.7	22.3	16.6	12.6		
	5%	WB	6.4	6.1	8.7	12.9	18.2	21.9	23.9	24.0	22.5	18.6	13.8	10.1		
		MCDB	7.6	7.9	11.4	16.0	21.8	25.7	27.8	27.1	24.6	20.7	15.2	11.3		
	10%	WB	4.7	4.5	7.1	11.4	16.8	21.0	23.2	23.3	21.5	17.3	12.2	7.8		
		MCDB	6.3	6.3	9.5	14.5	19.9	24.2	26.6	26.1	23.9	19.4	13.8	9.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.1	7.5	7.8	8.4	8.3	8.0	7.8	7.4	7.9	8.1	7.7	6.8		
		MCDBR	8.6	9.4	10.5	12.2	11.8	10.3	9.5	8.6	8.6	9.0	8.8	8.3		
	5% WB	MCWBR	7.7	7.3	6.9	6.4	5.8	4.6	3.9	4.0	4.8	6.2	7.4	8.0		
		MCWBR	8.5	8.8	9.7	10.5	10.6	8.9	7.9	7.0	6.7	7.8	7.7	8.1		
(40) Clear-Sky Solar Irradiance	taub		0.309	0.317	0.341	0.377	0.426	0.473	0.488	0.464	0.391	0.349	0.320	0.307		
	taud		2.481	2.461	2.444	2.378	2.272	2.179	2.148	2.206	2.381	2.514	2.547	2.542		
	Ebn at Noon		849	896	910	895	854	811	794	802	844	848	831	824		
	Edh at Noon		76	89	102	116	133	146	149	137	108	84	71	66		
(44) All-Sky Solar Radiation	RadAvg		1.85	2.72	3.74	4.80	5.36	5.86	5.98	5.32	4.36	2.96	2.06	1.55		
	RadStd		0.15	0.25	0.30	0.50	0.53	0.56	0.32	0.33	0.36	0.29	0.17	0.14		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+146	+80		
(47)	Regional (69 neighbors)	+0.39	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+108	+59		

Nomenclature: See separate page